# Lecture 27 — Midterm 2 review and synthesis

**Mon Nov 9**

Last updated: October 19, 2026.

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

Full review day for Midterm 2 (Friday, November 13). Related objectives: Lectures. Schedule.

## Prepare before class

1. Study guide — mark each outcome confident / shaky / blank.
2. Bring your definitions/formulas list and questions from Quizzes 5–9.
3. Start (or continue) the practice packet; we will work mixed items in class.

## Learning objectives

By the end of class, you should be able to:

1. Connect transformations, systems, dynamics, rank, inverses, and determinants.
2. Compare hand and computational solution methods.
3. Interpret software output instead of merely reporting it.
4. Solve a new application requiring more than one concept.
5. Identify gaps and make a targeted study plan before Friday.

## In class

- Quick scan of Unit II outcomes (matrices / $Ax$ / transforms $\rightarrow$ systems / RREF $\rightarrow$ dynamics $\rightarrow$ products / powers / cost $\rightarrow$ inverses / rank / determinants).
- Mixed practice from the packet (definitions, computations, harder items).
- Partner or board work on shaky topics from the study guide checklist.